Chapter 17

Safety Assessment

17.1 I now turn to outline the specific elements in the regime which I envisage, starting in this chapter with safety assessment. I will first describe the nature and value of safety assessment (paras 17.3-7), the models for its regulatory use furnished by the Safety Case in the Control of Industrial Major Accident Hazards (CIMAH) Regulations 1984 (paras 17.8-16) and by the arrangements in the Norwegian Continental Shelf (NCS) (paras 17.17-21), and the practice and intentions of the DEn (paras 17.22-28). Next I will state my views on, and proposals for, the role of safety assessment in the regime (paras 17.29-47). I will then consider the role of quantification and quantitative risk assessment (paras 17.48-61). I will conclude by considering the implications of my proposals for regulations, for the regime and for the regulatory body (paras 17.62-71).

17.2 The Piper disaster involved the realisation of potential major hazards. There was a leak and an explosion inside a module followed by the rupture of gas risers. Although Occidental could not but be aware of the existence of such hazards, it did not possess any system which ensured that such remote, but potentially disastrous, events were subjected to systematic scrutiny. There was for major projects no comprehensive system of safety assessment and management did not appear to appreciate fully the contribution which it could make. By contrast, the evidence showed that some companies, both those operating in the UKCS and in the NCS, require the formal use of safety assessment for major projects, and did so prior to Piper. The companies which gave evidence on this were clear that these activities were beneficial in the identification and control of hazards.

Formal safety assessment

17.3 Formal safety assessment (FSA) involves the identification and assessment of hazards over the whole life cycle of a project from the initial feasibility study through the concept design study and the detail design to construction and commissioning, then to operation, and finally to decommissioning and abandonment. The techniques used include hazard and operability (HAZOP) studies; quantitative risk assessment (QRA); fault tree analysis; human factors analyses; and safety audits. The need for FSA arises because the combinations of potential hardware and human failures are so numerous that a major accident hardly ever repeats itself. A strategy for risk management must therefore address the entire spectrum of possibilities.

17.4 In accordance with the usage of the main witnesses in this passage of evidence, I shall use the term "formal safety assessment", or FSA, to mean the process of assessment and "an FSA" to mean the output from this process and in particular an assessment essentially equivalent to a Safety Case. It is with this latter that I shall be primarily concerned in the first part of this chapter.

Current use of FSA offshore

17.5 Some companies operating in the North Sea already produce FSAs for major projects. Dr M S Hogh, Manager of Projects and External Affairs, Group Safety Centre, BP International, described the formal system used within BP. There is a formal Project Review Procedure conducted at 6 distinct stages in the course of a project, starting with definition and feasibility and going through to operation, in which independent audit teams seek to identify any outstanding safety issues. There is a formal requirement to carry out HAZOP studies at the detail design stage and the results are scrutinised by the audit team. FSAs were also described by Mr R E McKee, Chairman and Managing Director of Conoco (UK) Ltd, who stated that since Piper a separate group had been created to deal with this.

17.6 The value of an FSA to the company was illustrated by the Engineering Safety Plan for the Southern Basin Gas Development, or V Fields project, described by Mr M Ferrow, Manager, Safety and Quality Assurance, Conoco (UK) Ltd. The development involved a number of gas fields some 50 miles east of the Lincolnshire coast. The exercise was modelled on the Norwegian concept safety evaluation (CSE), a form of FSA carried out at the conceptual stage. The objectives were to demonstrate the safety and reliability of the design; to detail the operational requirements and limitations; and to provide the basis for continuing safety assurance after handover. One outcome of the work was a systematic, documented review of all significant accident scenarios and the associated precautions. Another was a lead in to the operating group of all the issues which the design group felt were important for safety. Some 500 operating practices were derived from the work. The documentation comprised in its detailed form some 11 volumes. The value of the plan was demonstrated, when, following Piper, the company reviewed its safety precautions. The availability of the plan documentation made this a relatively straightforward exercise.

17.7 As his own title indicated, Mr Ferrow regarded safety and quality assurance (QA) as linked. He described FSA as a subset of quality assurance. The in-house quality assurance system was used to ensure that findings of the V Fields study just described were properly closed out.

The CIMAH model

17.8 I now turn to consider the role of an FSA in the regulatory regime. The Inquiry heard of 2 existing models for this, the onshore Safety Case and the arrangements in the NCS. The Safety Case was described by Dr A D Sefton, a factory inspector based at Sheffield and leader of the HSE's Hazardous Installations and Transport of Dangerous Substances National Interest Group.

17.9 Onshore major hazard installations are subject to the CIMAH Regulations. Reg 7 requires that the operator should provide the HSE with a written report on the safety of the installation. The report is commonly called the Safety Case. These regulations had their origins in the Flixborough disaster in 1974 and the work of the Advisory Committee on Major Hazards (ACMH) and in the Seveso disaster. They effect, and are confined to, the implementation of the EC Directive on Major Accident Hazards, the so-called "Seveso" Directive. They require demonstration of safe operation (Reg 4), notification of major accidents (Reg 5), a written report (the Safety Case) (Reg 7), updating of the report (Reg 8), an obligation to supply the HSE with further information (Reg 9), preparation of an on-site emergency plan (Reg 10) and provision of information to the public (Reg 12). There are also requirements on the local authority to prepare an off-site emergency plan (Reg 11).

17.10 The contents of the written report are specified in Schedule 6 of the regulations. The 4 main headings relate to information on every dangerous substance involved in the activity; on the installation itself; on the management system; and on the potential major accidents. The information required on the management system includes the staffing arrangements; the arrangements made to ensure that the means provided for safe operation are properly designed, constructed, tested, operated, inspected and maintained; and the arrangements for training. That required on the potential major accidents includes the potential sources of a major accident and the conditions or events which could be significant in bringing one about; the features of the plant which are significant as regards potential for a major accident or its prevention or control; the measures taken to prevent, control or minimise the consequences of a major accident; and the emergency procedures.

17.11 In the first instance the Safety Case is a means by which an operator demonstrates to itself the safety of its activities. The value of such a demonstration

was illustrated in the evidence of Mr Ferrow concerning an FSA which I have already mentioned.

17.12 The Safety Case also serves as the basis for the regulation of major hazard activities, as described by Dr Sefton. Existing major hazard installations were already subject to the Health and Safety at Work etc Act (HSWA) and even before the CIMAH Regulations had been the subject of attention by the local inspector. A CIMAH site is normally visited annually. On receipt of a Safety Case the HSE first checks to ensure that all the information required is provided and to identify any matter of immediate concern. The report is then assessed by a multi-disciplinary team including specialists from HSE's Technology Division, the local area inspectors and as necessary local specialists in the Field Consultant Groups. Any matters of concern are then taken up by letter or by visit. Following this initial response, the report constitutes an important input into the inspection strategy and provides a basis for selecting areas which should receive priority attention. Examples cited by Dr Sefton were the use and maintenance of an item of hardware such as pressure relief devices or a procedural matter such as operating instructions. In the course of in-depth inspection of these items inspectors would always test management and organisation of the installation by reference to any failings detected. Many operators have reported that they found the exercise of producing a Safety Case valuable. Often it would be the first time that a report had been made of the major hazard aspects of the installation. Many stated that the exercise had led them to make changes in their approach and improvements to systems and procedures. Dr Sefton was at pains to point out that the Safety Case was not a licensing or approval system, which might be thought to transfer some of the responsibility to the licensing authority. He was not even sure that the HSE "accepted" the report. What it did was to satisfy itself that the information provided complied with that required and then use that information in its inspection; if there were any serious concerns arising out of the report, it would take them up.

17.13 With regard to the level of expertise required within the company to prepare the Safety Case, the guidance notes state:

> "A partial answer is to suggest that if a manufacturer was unable to meet most if not all of the aims of the Safety Case set out in para 106 by using his own staff, doubts would arise about his competence to manage a major hazard activity ..." (para 114).

In practice Safety Cases submitted are for the most part prepared by the operator's personnel, although some use is made of consultants for specialist work such as consequence modelling, particularly by smaller companies. In assessing the Safety Case, the HSE is able to bring to bear the full range of expertise required. It possesses this expertise in-house. However, it does make use of consultants to assist with peak workloads.

17.14 The Safety Case is concerned with management and software as well as with hardware, as indicated by the information on management and management systems required in para 4 of the Schedule. Amplifying these requirements Dr Sefton stated:

> "Information should be given which details operation and revision of safety policies; the setting or adoption of design and construction standards; quality assurance arrangements, operating procedures, training, management supervision, monitoring, staff welfare and management structure. All these separate elements are necessary to describe fully a system of management control and the report should give some indication of activity within every element. Control in the above spheres of activity requires:
>
> (a) identification of work required to achieve the desired objectives; (b) the establishment of standards for described activities; (c) performance measurement to assess the degree of compliance with standards; (d) evaluation of performance